

Gaze-Glide (Eye-Gaze Controlled Wheelchair)

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Table of Contents

- ➊ Introduction
- ➋ Literature Review
- ➌ Objectives
- ➍ Architecture
- ➎ Application
- ➏ Methodology
- ➐ Design Formulation
- ➑ Project Phase 1
- ➒ Project Phase 2
- ➓ Expected Results
- ➙ References

Introduction

- **Overview of Eye-Gaze Controlled Wheelchair Technology**
 - Utilizes eye movement to control wheelchair direction and speed.
 - Integrates computer vision and machine learning for accurate gaze detection.
- **Importance and Relevance for ALS Patients**
 - Provides mobility independence for individuals with limited physical abilities.
 - Enhances quality of life by enabling autonomous navigation.
- **Objectives of the Study**
 - Develop a reliable and efficient eye-tracking system.
 - Ensure user-friendly interface and seamless integration with wheelchair hardware.

Literature Review - Part 1

- **Development in Autonomous and Assistive Wheelchair Control**
 - Review of traditional control methods: manual and joystick-based systems.
 - Advances in autonomous systems for enhanced safety and navigation in complex environments.
- **Comparative Analysis of Control Modalities: Gesture, Voice, BCI, and Eye-Tracking**
 - Examination of gesture-based controls and their reliability in noisy or dynamic conditions.
 - Voice control systems: accuracy, usability challenges, and response times.

Literature Review - Part 2

- **Comparative Analysis of Control Modalities: BCI and Eye-Tracking (cont.)**
 - Brain-Computer Interface (BCI) controls: technical complexity, accuracy, and cognitive load on users.
 - Eye-tracking systems: gaze detection accuracy, real-time responsiveness, and suitability for users with varying needs.
- **Technical Challenges and Practical Constraints**
 - Signal interference, variability in environmental conditions, and associated user fatigue in continuous use.
 - Scalability limitations, adaptability to diverse user requirements, and user-specific customization.
- **Recent Innovations in Eye-Tracking for Assistive Devices**
 - Exploration of infrared-based eye-tracking technologies to enhance gaze recognition in low-light and variable lighting conditions.
 - Development of algorithms to distinguish intentional commands from accidental gaze movements.

Objectives

- **To Design a Highly Accurate Eye-Gaze Controlled Wheelchair**
 - Achieve precise detection of eye movements under various conditions.
 - Minimize error rates in command recognition.
- **To Develop a Smooth Motion Acceleration Model**
 - Implement algorithms that ensure gradual acceleration and deceleration.
 - Enhance user safety by preventing abrupt movements.
- **To Enhance User Safety and Comfort**
 - Integrate safety features such as obstacle detection and automatic braking.
 - Design user-friendly interfaces for ease of use and reduced fatigue.

Architecture

Professional System Architecture Block Diagram Eye-Cotrolled Wheelchair

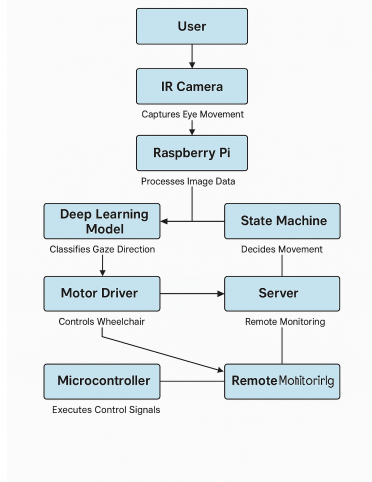


Figure: System Architecture

Architecture

The eye-controlled wheelchair system consists of the following main components:

- 1. **Miniature Electric Wheelchair**
- 1. **Vision System**
- 1. **Main Control System**

The system captures facial and eye images through the IR camera, processes them with a neural network, and translates gaze movements into directional commands for wheelchair control:

- **Neural Network Model:** A custom deep learning model, trained for gaze detection, estimates the gaze direction accurately based on IR camera input.

Architecture (cont'd)

A specialized dataset, compiled from various environments, is used to train the gaze detection model:

- **Dataset Collection:** Eye-tracking data is collected in both real-world and controlled conditions for diverse lighting and background scenarios.
- **Model Training:** The neural network is trained with convolutional and attention-based layers for reliable gaze detection and direction classification.

The system combines various components for seamless, real-time control:

- **Hardware Components:** An IR camera for gaze detection and a control system based on Raspberry Pi.
- **Software Platform:** Python-based control scripts, potentially utilizing libraries for real-time tracking and gaze interpretation.
- **Servo Mechanism:** Connected to the wheelchair control system, allowing smooth, directional movement based on gaze input.

Application

- **Medical and Assistive Applications for ALS Patients**
 - Provides autonomy in mobility for individuals with advanced ALS.
 - Facilitates daily activities and enhances independence.
- **Potential Uses in Elderly Care and Rehabilitation**
 - Assists elderly individuals with limited mobility in navigating their environment.
 - Supports physical therapy by enabling controlled movement exercises.
- **Future Expansion into Other Disabilities**
 - Adaptable for users with different types of motor impairments.
 - Potential integration with other assistive technologies for comprehensive support.

Methodology

- **Objective Definition:**

- Develop a cost-effective, miniature eye-controlled wheelchair model using gaze detection for navigation.
- Prioritize safety, ease of use, and operability in varied lighting conditions.

- **Data Collection and Model Training:**

- Collect eye movement data (directions like left, right, up, down) in both simulated and real environments, using an IR camera.
- Train a Convolutional Neural Network (CNN) with attention mechanisms to accurately interpret gaze directions, accounting for different lighting conditions.

- **Gaze Detection and Signal Processing:**

- Deploy the trained model on a Raspberry Pi for real-time gaze detection.
- Process gaze signals to generate movement commands, translating user gaze into specific directional controls (e.g., forward, backward, left, right).

Methodology (cont'd)

- **Wheelchair Control and Navigation:**

- Interface the gaze-based movement commands with the motor driver to control the wheelchair's motion.
- Implement safety protocols, such as obstacle detection and smooth motion control, to ensure safe navigation.

- **System Testing and Optimization:**

- Conduct trials in different environments to assess system responsiveness, accuracy of gaze detection, and reliability of movement.
- Refine model parameters and control mechanisms to enhance system robustness and user safety.

Design, Simulation, and Prototype Development

- **System Architecture:**
 - Develop a block diagram outlining interactions between components.
- **Software Design:**
 - Plan algorithms for gaze detection and motor commands.
 - Design data flow for processing gaze input.
- **Prototype Development:**
 - Assemble chassis, connect motors and camera.
 - Set up wiring and install camera.

Testing, Optimization, and Future Enhancements

- **Testing and Validation:**

- Test each command (move, stop, turn) and validate gaze detection.
- Conduct user testing and gather feedback.

- **Optimization:**

- Refine gaze detection and motor control.
- Implement additional safety features.

- **Future Enhancements:**

- Consider obstacle avoidance or autonomous navigation.
- Plan potential scaling to a full-sized wheelchair.

System Model and Components

- **Objective:** Control wheelchair using eye movements and blinks.
- **Components:**
 - **Camera:** Captures eye movements.
 - **Microcontroller:** Processes eye data, controls motors.
 - **Motors & Drivers:** Drive wheels based on input.
 - **Battery:** Powers system.

Control Algorithms

- **Movement Commands:**

- Forward: Gaze centered.
- Left/Right: Gaze in respective direction.
- Stop: Prolonged blink or specific pattern.

Design Flow and User Interface

- **System Flow:**

- Eye image capture → Gaze detection → Command processing → Motor control.

- **User Interface:** Displays real-time movement status and feedback.

Performance Metrics

- **Responsiveness:** Measure delay in wheelchair response to gaze.
- **Accuracy:** Calculate rate of correct gaze classifications.

1. Project Phases

- **Phase 1: Define Objectives** (Weeks 1–2)
 - Define objectives, research background, establish requirements.
- **Phase 2: Component Selection and Procurement** (Weeks 3–4)
 - Select microcontroller, gather hardware components.
- **Phase 3: Design and Simulation** (Weeks 5–6)
 - Develop system architecture, design control algorithms.
- **Phase 4: Prototype Development** (Weeks 7–8)
 - Assemble and configure hardware setup.
- **Phase 5: Project Planning and Research** (Weeks 9–10)
 - Conduct testing, optimize, and document.

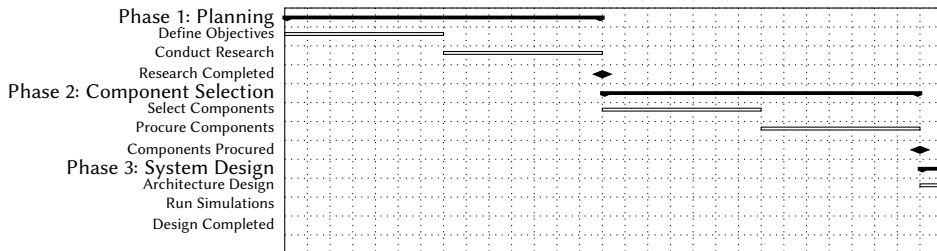
Detailed Schedule

Phase	Tasks	Duration	Timeline
Phase 1	Define objectives, research	2 weeks	Weeks 1–2
Phase 2	Select components, procurement	2 weeks	Weeks 3–4
Phase 3	Design architecture, simulation	2 weeks	Weeks 5–6
Phase 4	Prototype assembly, wiring	2 weeks	Weeks 7–8
Phase 5	Project Planning and Research, wiring	2 weeks	Weeks 9–10

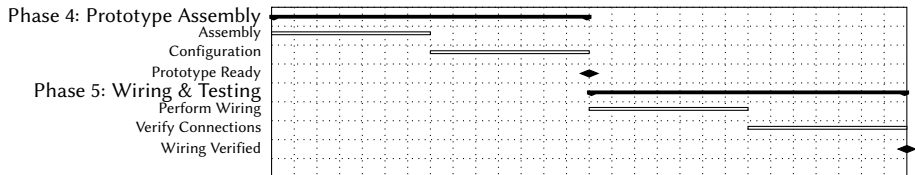
Milestones

- **End of Week 2:** Objectives and research completed.
- **End of Week 4:** Components selected and procured.
- **End of Week 6:** System design and simulation finished.
- **End of Week 8:** Prototype assembly and configuration completed.
- **End of Week 10:** After successfully wiring all necessary components and verifying connections.

Project Timeline (Part 1)



Project Timeline (Part 2)



Resources and Tasks by Project Phase

Phase	Human Resources	Material Resources	Tasks
Project Planning and Research	Project Manager, Hardware Engineer	-	Define objectives, conduct background research, establish requirements, assign roles
Component Selection and Procurement	Hardware Engineer	Microcontroller, camera, motors, chassis	Select microcontroller, procure components, ensure compatibility
Design and Simulation	Project Manager, Software Developer, Machine Learning Specialist	Development Environment	Create system architecture, design control algorithms, simulate gaze detection, plan safety features

Resources and Tasks by Project Phase

Phase	Human Resources	Material Resources	Tasks
Prototype Development	Hardware Engineer	Chassis, motors, motor drivers, power supply	Assemble chassis, position camera, connect wiring, initial power testing

2. Project Phases

- **Phase 5: Component Selection and Procurement** (Weeks 9–10)
 - Select microcontroller, gather hardware components.
- **Phase 6: Model Training and Optimization** (Weeks 11–12)
 - Train the eye-tracking model using collected datasets.
- **Phase 7: Real-Time Processing and Response Tuning** (Weeks 13–14)
 - Develop real-time gaze tracking for controlling wheelchair movement.
- **Phase 8: Hardware Integration and Initial Testing** (Weeks 15–16)
 - Deploy the trained model on Raspberry Pi or an edge computing device.

Detailed Schedule

Phase	Tasks	Duration	Timeline
Phase 6	Component Selection and Procurement	2 weeks	Weeks 11–12
Phase 7	Model Training and Optimization	2 weeks	Weeks 13–14
Phase 8	Real-Time Processing and Response Tuning	2 weeks	Weeks 15–16
Phase 9	Hardware Integration and Initial Testing	2 weeks	Weeks 17–18

Milestones

- **End of Week 12:** All necessary hardware components acquired and verified.
- **End of Week 14:** Eye-tracking model trained with optimized accuracy.
- **End of Week 16:** Model integrated with real-time processing for smooth wheelchair control.

Project Timeline (Part 1)

Phase 6: Component Selection

Select Components

Procure Components

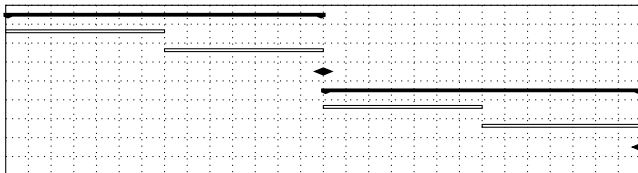
Components Procured

Phase 7: Model Training

Train Initial Model

Optimize Model

Model Optimized



Project Timeline (Part 2)

Phase 8: Real-Time Processing

Implement Real-Time System

Tune Processing Speed

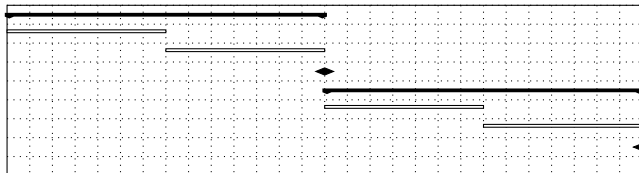
Real-Time Tuning Completed

Phase 9: Hardware Integration

Integrate Hardware

Initial Testing

Hardware Testing Completed



Resources and Tasks by Project Phase

Phase	Human Resources	Material Resources	Tasks
Component Selection and Procurement	Hardware Engineer	Microcontroller, sensors, additional hardware	Select microcontroller, gather hardware components, ensure compatibility.
Model Training and Optimization	Machine Learning Specialist, Software Developer	Labeled datasets, training environment, computing resources	Train the eye-tracking model using collected datasets, optimize accuracy and response time.
Real-Time Processing and Response Tuning	Software Developer, Embedded Systems Engineer	Raspberry Pi, eye-tracking camera, real-time processing tools	Develop real-time gaze tracking system, fine-tune processing for wheelchair movement control.
Hardware Integration and Initial Testing	Hardware Engineer, Embedded Systems Engineer	Raspberry Pi or edge computing device, wheelchair hardware	Deploy trained model on Raspberry Pi or edge device, integrate with wheelchair hardware, conduct initial tests.

Expected Results

- **Functional Outcomes**

- Accurate Eye-Gaze Control:
 - Move forward with a straight gaze.
 - Turn left/right with respective gaze directions.
 - Stop with a predefined blink pattern or prolonged blink.
- Smooth and Responsive Movement:
 - Minimal delay in commands, ensuring a smooth and safe ride.

- **System Performance Metrics**

- Gaze Detection Accuracy: Achieve at least 90% accuracy to ensure reliable control.
- Response Time: Less than 500 ms from gaze input to movement.
- Battery Life: Minimum of 4 hours of continuous operation.

- **User Experience and Usability**

- Ease of Use: Intuitive controls with a learning curve of less than 10 minutes.
- User Satisfaction: Aim for an 85% satisfaction rating based on control accuracy and ease of learning.
- Accessibility: Ensure usability for individuals with limited physical movement.

- **Documentation and Future Development**

- Comprehensive Documentation: Record design, testing, and feedback.
- Performance Analysis: Report on metrics met and areas for improvement.
- Future Work: Recommend enhancements, such as advanced obstacle detection or improved gaze tracking.

References

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Thank You!
any questions